L6. 17/09/2020

The following statements are equivalent $\quad A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$

(a) The controllability matrix $\mathcal{C}_{AB}$ has full row-rank, i.e. $\text{rank}(\mathcal{C}_{AB}) = n$

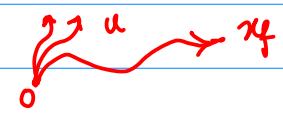
(b) (A,B) is controllable

(c) The matrix $[A-\lambda I \quad B]$ has full row-rank $\forall \lambda \in \mathbb{C}$ (PBH-test)

✓ (d) The matrix $W_t := \int_0^t e^{A\tau} B B^* e^{A^*\tau} d\tau$ is +ve def^n for any $t \in \mathbb{R}$.

(e) Let $\lambda \in \sigma(A)$ and x be it's corresponding left eigenvector. (Then $x^* A = x^* \lambda$)
Then $x^* B \neq 0$. $\qquad x^*[A-\lambda I \quad B] = [0 \quad x^* B]$

(f) The eigen values of $(A+BF)$ can be freely assigned by suitable F
* (with the restriction that complex eigenvalues of $A+BF$ appears as conjugate pair.)



$B \to b$, $n\times 1$

SISO

Suppose (A,b) is a controllable pair. Then their exists a non-singular $T \in \mathbb{R}^{n\times n}$ such that

Controller Canonical Form (CCF)

$$\tilde{A} := T^{-1} A T = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \qquad \tilde{b} := T^{-1} b = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix},$$

where $\chi_A(s) = s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0$

Proof: $\chi_A(s) = s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0$

C.H: $A^n + a_{n-1} A^{n-1} + \cdots + a_1 A + a_0 I = 0$

$\Rightarrow A^n = -a_{n-1} A^{n-1} - a_{n-2} A^{n-2} + \cdots - a_{n-1} A - a_0 I$

$A = -1$

$\dot{x} = -x + u$

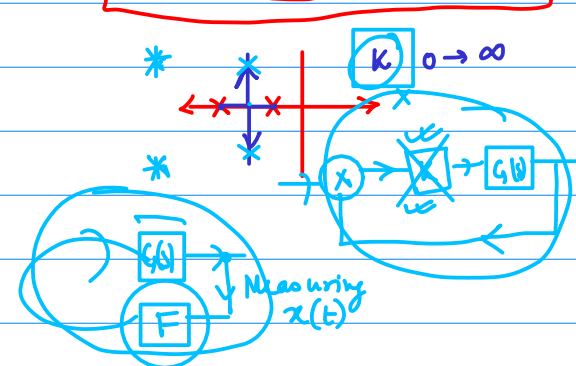
$x(t) = e^{-t} x_0 + \int e^{-(t-\tau)} u(\tau) d\tau$

x_f

$x(t) = e^{-2t} x_0$

$$\tilde{A}\,[b \quad Ab \cdots A^{n-1} b] = [Ab \quad A^2 b \cdots A^{n-1} b \quad A^n b]$$

$$[\quad Ab \cdots A^{n-1} b \quad -a_{n-1} A^{n-1} b - a_{n-2} A^{n-2} b - \cdots - a_0 b \quad]$$



$$A\underbrace{\begin{bmatrix} b & Ab & \cdots & A^{n-1}b \end{bmatrix}}_{P} = \begin{bmatrix} Ab & A^2b & \cdots & A^{n-1}b & -a_{n-1}A^{n-1}b - a_{n-2}A^{n-2}b - \cdots - a_0 b \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} b & Ab & \cdots & A^{n-1}b \end{bmatrix}}_{P} \underbrace{\begin{bmatrix} 0 & 0 & \cdots & \cdots & -a_0 \\ 1 & 0 & \cdots & \cdots & -a_1 \\ 0 & 1 & \cdots & \cdots & -a_2 \\ \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & \cdots & -a_{n-2} \\ 0 & 0 & \cdots & \cdots & -a_{n-1} \end{bmatrix}}_{\hat{A}}$$

$$b = \underbrace{\begin{bmatrix} b & Ab & \cdots & A^{n-1}b \end{bmatrix}}_{P} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \rightarrow ③$$

$$AP = P\hat{A} \Rightarrow \boxed{P^{-1}AP = \hat{A}} \longrightarrow ①$$

$$\chi_{\tilde{A}}(s) = s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0$$

$$\boxed{\det(sI - A) = \det(sI - \hat{A})}$$

$$\tilde{A}\underbrace{\begin{bmatrix} \tilde{b} & \tilde{A}\tilde{b} & \cdots & \tilde{A}^{n-1}\tilde{b} \end{bmatrix}}_{W} = \underbrace{\begin{bmatrix} \tilde{b} & \tilde{A}\tilde{b} & \cdots & \tilde{A}^{n-1}\tilde{b} \end{bmatrix}}_{W} \underbrace{\begin{bmatrix} 0 & 0 & \cdots & \cdots & -a_0 \\ 1 & 0 & \cdots & \cdots & -a_1 \\ 0 & 1 & \cdots & \cdots & -a_2 \\ \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & \cdots & -a_{n-2} \\ 0 & 0 & \cdots & \cdots & -a_{n-1} \end{bmatrix}}_{\hat{A}}$$

$$\tilde{A}W = W\hat{A} \Rightarrow \boxed{W^{-1}\tilde{A}W = \hat{A}} \rightarrow ②$$

if & only if $(\tilde{A}, \tilde{b})$ cont^r

$$W^{-1}\tilde{A}W = P^{-1}AP$$

$$\Rightarrow WP^{-1}A\underbrace{\boxed{PW^{-1}}}_{T} = \tilde{A} \Rightarrow T^{-1}AT = \tilde{A}$$

$$\tilde{b} = \begin{bmatrix} \tilde{b} & \tilde{A}\tilde{b} & \cdots & \tilde{A}^{n-1}\tilde{b} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \end{bmatrix}$$

$$③ \& ④ \Rightarrow W^{-1}\tilde{b} = P^{-1}b$$

$$\Rightarrow \boxed{\tilde{b} = WP^{-1}b} \quad ④$$

$$\dot{x} = Ax + Bu \qquad T$$

$$\sigma(T^{-1}AT) = \sigma(A)$$

$$x(t) = T\hat{x}(t)$$

$$T\dot{\hat{x}}(t) = AT\hat{x}(t) + Bu \Rightarrow \underbrace{\dot{\hat{x}}(t) = T^{-1}AT\,\hat{x}(t) + T^{-1}B\,u(t)}$$

3x3 example

$$\dot{\hat{x}}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \hat{x}(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$u(t) = F\hat{x}(t) \qquad F \in \mathbb{R}^{1\times n}$$

$$F := \begin{bmatrix} f_1 & f_2 & f_3 \end{bmatrix} \qquad f_1, f_2, f_3 \in \mathbb{R}$$

$$\dot{\hat{x}}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \hat{x}(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} f_1 & f_2 & f_3 \end{bmatrix} \hat{x}(t)$$

$$= \left(\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ f_1 & f_2 & f_3 \end{bmatrix} \right) \hat{x}(t)$$

$$= \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0+f_1 & -a_1+f_2 & -a_2+f_3 \end{bmatrix}}_{A+BF} \hat{x}(t)$$

$$\chi_{A+BF}(s) = s^3 + (a_2 - \underline{\underline{f_3}})s^2 + (a_1 - \underline{\underline{f_2}})s + (a_0 - \underline{\underline{f_1}})$$

e.g. $A = \begin{bmatrix} 0 & 1 \\ -4 & -4 \end{bmatrix}$ $\quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\quad \chi_A(s) = s^2 + 4s + 4 = (s+2)^2 \quad \sigma(A) = \{-2, -2\}$

$\underline{\underline{\{-1, -3\}}}$

$F = \begin{bmatrix} f_1 & f_2 \end{bmatrix}$

$-1+j, -1-j$

$$A + BF \quad \begin{bmatrix} 0 & 1 \\ \underbrace{-4+f_1}_{-3} & \underbrace{-4+f_2}_{-4} \end{bmatrix} \qquad \chi_{A+BF}(s) = (s+1)(s+3)$$

$$= s^2 + \underset{\uparrow}{4}s + \underset{\uparrow}{3}$$

$$-4 + f_1 = -3 \Rightarrow f_1 = 1, \quad -4 + f_2 = -4, \quad f_2 = 0$$

$$F = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

After feedback: $\tilde{u}(t) = \tilde{F}\tilde{\hat{x}}(t)$ $\qquad \dot{\hat{x}}(t) = (A + B\tilde{F})\hat{x}(t)$

$$\boxed{\hat{x}(t) = e^{(A+BF)t}\hat{x}(0)}$$